# Chandrashekhar Choudhary

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## SUMMARY

Ph.D. candidate in Electrical Engineering specializing in semiconductor device research, MEMS, sensor design, and microfabrication. Extensive experience with sensor and system design, advanced materials characterization, and cleanroom fabrication. Proven ability to lead experimental R&D projects, collaborate across multi-functional teams, and communicate findings through technical reports and publications.

## EDUCATION

**Ph.D in Electrical Engineering (Sensor and System Design Track)** — *Jun. 2026*
*University of Cincinnati (UC), Cincinnati, Ohio* — *GPA: 4.0/4.0*

**Master of Technology (Electronics System Design)** — *July. 2018*
*International Institute of Information Technology - Bangalore (IIIT Bangalore), Bangalore, India* — *GPA: 3.7/4.0*

## GRADUATE RESEARCH EXPERIENCE

**MEMS based Acoustophoretic Nanoparticle Separator** — ***Present***
- Engineered a two-stage microfluidic separator for the precise separation of airborne nanoparticles using acoustophoretic principles.
- Performed design simulations and optimizations using COMSOL and MATLAB to enhance device performance and separation efficiency, device fabrication and implementation in progress.

**Design and Packaging of Implantable Biomedical Device for Osteoarthritis Monitoring** — ***Present***
- Engineered passive, wireless, implantable biosensor using micro EDM for in vivo fluid analysis in OA patients.
- Fabricated a micro-scale biocompatible polymer package with integrated Nitinol anchors for sensor deployment using a mold fabricated with EDM
- Developed comprehensive process flow to seamlessly integrate the sensor and package, ensuring functionality and performance.

**High-Efficiency Miniature Multi-Tip Corona Charger for Nanoparticle Sensor** — *2023-25*
- Designed and optimized a novel micromachined indirect unipolar corona charger with multiple discharge sites for efficient nanoparticle detection.
- Conducted materials characterization and device analysis using microscopy, surface profilometry, and electrical metrology tools.
- Published experimental results demonstrating **<3.25%** particle loss at **10.55 nm;** optimized for batch fabrication with thermo-compression bonding.

**Battery-Powered Wireless Sensor Network** — *2021-22*
- Architected low-power wireless network for continuous monitoring using acoustic and novel calorimetric sensors.
- Programmed embedded devices for data acquisition and optimized wireless transmission, validated through 30-day field testing.

**Non-Invasive Calorimetric Sensor for Waterflow Event Detection in Premise Plumbing Systems** — *2021-22*
- Engineered a novel thermal isolation-based non-invasive calorimetric sensor for ultra-low signal (**0.1ºC**) detection of waterflow events.
- Utilized microfabrication and material optimization techniques to maximize system sensitivity and reliability.
- Seamless integration of the designed sensor with the battery-powered wireless sensor network system.

**Development of Silicon-Based Piezoresistive Pressure Sensors** — *Spring'21*
- Fabricated pressure sensors on silicon wafers utilizing photolithography, ion implantation, metallization, and e-beam evaporation techniques.
- Characterized device properties using scanning electron microscopy (SEM) and resistance-voltage measurements.
- Achieved **~90%** device accuracy after process optimization and rigorous testing in cleanroom settings.

## WORK EXPERIENCE

**Cisco Systems** — **Bangalore, India**
*Software Engineer – G8(3)* — *Aug. 2018 – Dec. 2020*
- Led Cisco's enterprise switching software development, enhancing programmability, automation, and tooling for 50M+ devices globally.
- Delivered **10+** features across **6** releases on IOS-XE software powering **50M+** switches and routers.
- Led cross-functional effort to build an internal automated testing platform → reduced org-wide inefficiencies by **97%.**
- Resolved **50+** escalated customer issues through deep diagnostics and stable patch delivery.
- Built engineer utilization model adopted org-wide → boosted labor productivity by **25%.**

*Software Engineer Intern* — *Jan. 2018 – June 2018*
- Developed RF profiling application for client fingerprinting before onboarding onto wireless networks.
- Integrated security solution with Cisco routers to enhance network security.

*Software Engineer Intern* — *May. 2017 – July 2017*
- Created web-based dashboard to analyze and visualize customer bug data across Cisco components.
- Enabled data-driven decision making for addressing high-bug components.

**Ingersoll Rand** — **Bangalore, India**
*Hardware Intern* — *Jan. 2016 – June 2016*
- Developed hardware simulator and firmware to emulate sensor signals for testing ThermoKing SR4 controller boards.

**Schnieder Electric** — **Bangalore, India**
*Hardware Intern* — *Jun. 2015 – July 2015*
- Created multi-protocol gateway using Raspberry Pi for building automation applications (Modbus/BACnet).

## SKILLS

- **Semiconductor Fabrication:** Micro-EDM, Photolithography, Oxidation, Metallization, Etching, E-beam evaporation, Cleanroom process optimization, 3D Printing, Microfabrication, Polymer Molding & Casting, Electropolishing, Dip Coating
- **Characterization Tools:** SEM, Optical Microscopy, X-ray imaging, Surface Profilometry, Electrical Metrology, Schlieren imaging
- **Embedded Systems:** FreeRTOS, GDB, Buildroot, Linux device drivers, Firmware development, RTOS concepts, Baremetal programming, QT
- **Softwares:** SolidWorks, COMSOL Multiphysics, MATLAB, Eagle, LabVIEW, NI Multisim, Origin
- **Programming:** Python, C, C++
- **Interpersonal Skills:** Mentorship, Teamwork, Problem-Solving, Technical reports, Cross-functional collaboration, Presentations